

Exp 4 Design of Low Pass Filter using Windowing Techniques

4.1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique

Program:

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');

disp(b_rect);

%% --- 2. FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%% --- 3. FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann);

%% --- Plot Frequency Responses ---
```

```
figure;  
  
freqz(b_rect,1);  
  
title('Low-pass FIR using Rectangular Window');  
  
figure;  
  
freqz(b_hamm,1);  
  
title('Low-pass FIR using Hamming Window');  
  
figure;  
  
freqz(b_hann,1);  
  
title('Low-pass FIR using Hanning Window');
```

Outputs:

```
--- Rectangular Window Coefficients ---  
0.0126  
0  
-0.0137  
0  
0.0150  
0  
-0.0166  
0  
0.0185  
0  
-0.0210  
0  
0.0242  
0  
-0.0286  
0  
0.0349  
0
```

```

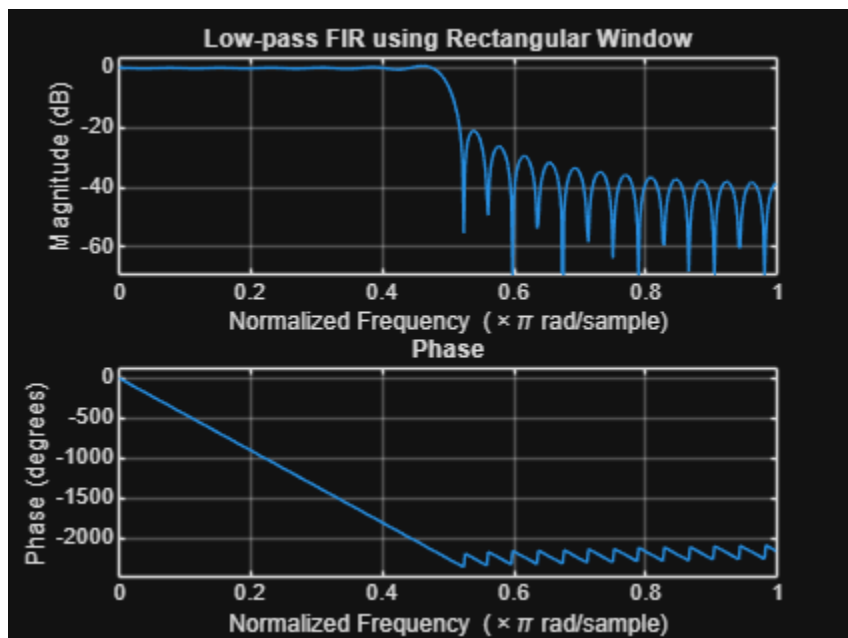
--- Hamming Window Coefficients ---
0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125
0
-0.0181
0
0.0260
0

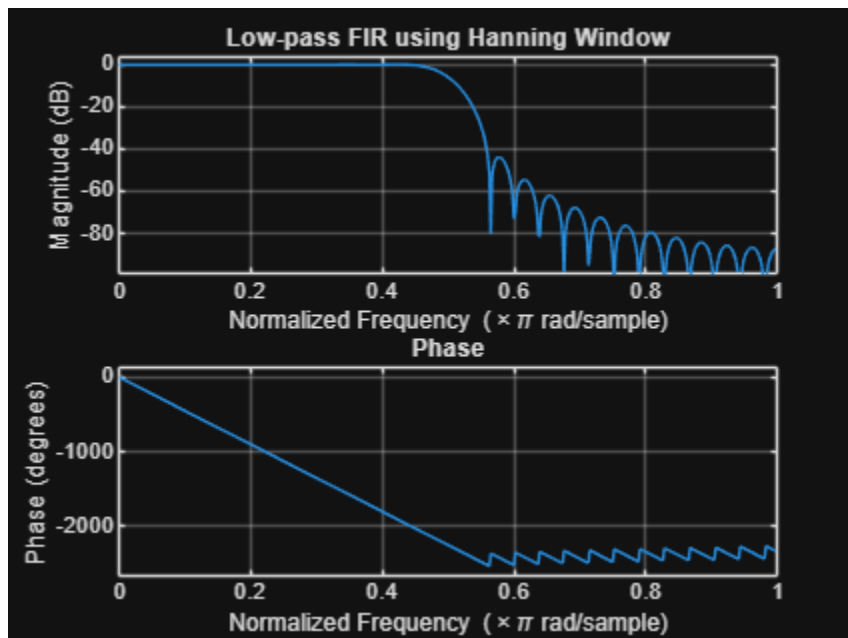
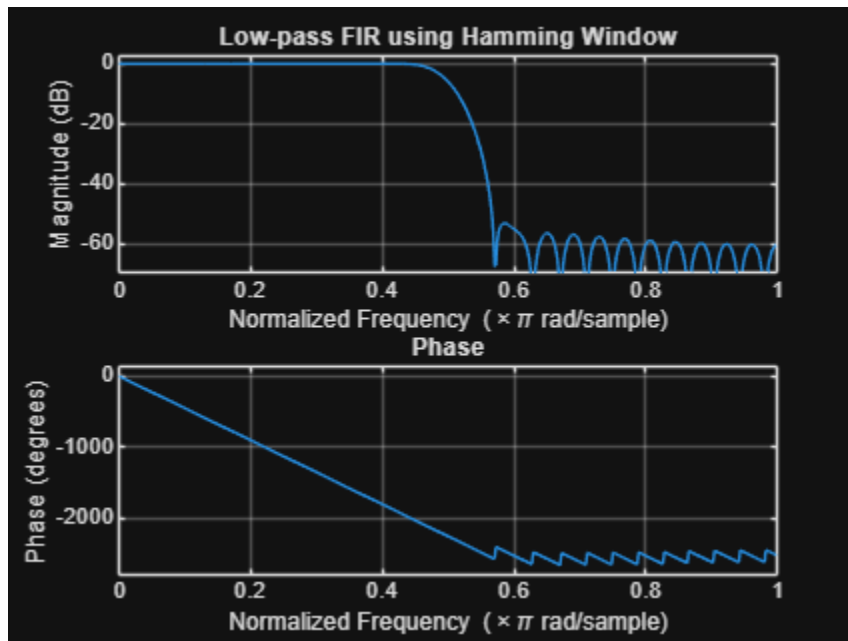
```

```

--- Hanning Window Coefficients ---
0.0000
0
-0.0004
0
0.0013
0
-0.0028
0
0.0050
0
-0.0081
0
0.0122
0
-0.0179
0
0.0259
0

```





4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

Program:

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm);

%% --- Plot Frequency Responses ---

figure;
```

```
freqz(b_hamm,1);
```

```
title('Low-pass FIR using Hamming Window');
```

Outputs:

```
--- Hamming Window Coefficients ---  
0.0010  
0  
-0.0013  
0  
0.0021  
0  
-0.0034  
0  
0.0055  
0  
-0.0084  
0  
0.0125  
0  
-0.0181  
0  
0.0260  
0
```

